BT 503 Quiz No. 1 Environmental Biotechnology (01-12-2022)

Sr. No.	Questions	Answers
1	Which of the following statements about xenobiotics is FALSE?	They are a sub class of Antibiotics
2	Biological treatments are not best suited to use in which types of soil?	All
3	Which component is not a xenobiotic?	Water
4	Why Extremozymes are important?	Because the production extremozymes is more industrial level
5	Halophiles are extremely important in environmental biotechnology. These are microorganisms that survive in highly environment. They do so by ensuring that.	Their cytoplasm contains a higher solute concentration than is present in their surroundings.
6	Which of the following is not a intervention of environmental biotechnology	Antibiotic development
7	The theory of evolution by natural selection was independently developed by	Charles Darwin and Alfred Wallace
8	To design production systems which avoid the potential for environmental contaminations in the challenge of	Green Chemistry
9	Define a Methanogen?	That produce methane as a metabolic byproduct in hypoxic conditions
10	Name the process by which land resources are restored to their former state or baseline condition is known as	Land remediation
11	Bacteria have also adapted to degrade man-made organics called	Xenobiotics
12	Find out the extreme Halophile among the organisms below	Salinibacter ruber
13	The existence of microbes capable of surviving environments has been known since the	1960s
14	Pyrolobus fumarii, optimum temperature for reproduction is	105-C
15	The theory of evolution by natural selection was independently developed by	Charles Darwin and Alfred Wallace
16	Many surface structural proteins in halophiles require severely elevated concentrations of	Sodium Salts
17	Microbes and enzymes typically function at	Low temperature, low pressure
18	What is Thermus Aquaticus?	Gram Negative Bacteria
19	Typically, the soil types best suited to biotechnological interventions are	Sands and gravels

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20	In order to metabolize crude oil, first patent for which genetically modified bacterial strain was granted in 1980	**Pseudomonas aeruginosa**
21	Name any endocrine Disruptions?	**Dioxins**
22	Find out the extreme Halophile among the organisms below	**Salinibacter ruber**
23	The first working draft of the human genome sequence was published in	**2000**
24	What are GMOs?	**Genetically Modified Organisms**
25	Species found in the deep ocean volcanic vents knowns as Black Smokers	**All**
26	Typically, the soil types best suited to biotechnological interventions are	**Sands and gravels**
27	Which term explains Biomethane best?	**Sustainable energy**
28	Eco-friendly means	**Earth friendly**
29	What is the estimative growth of global environmental market	**74 billion US dollars**
30	Which statement about endocrine disruption is TURE?	**Ferminisation of male fish**
31	Bacteria have also adapted to degrade man-made organics called	**Xenobiotics**
32	The estrogen is degraded completely to	**Carbon dioxide and water**
33	To design production systems which avoid the potential for environmental contaminations in the challenge of	**Green chemistry**
34	Bioaugmentation is	**The introduction more archaea bacteria**
35	Water treatment occurs for _____% of the total	**25**
36	Alkaliphiles are naturally occurring species, typically enduring PH of	**PH 9**
37	Which of the following statement is FALSE?	**Psychrophiles are found in regions with extremely high temperatures**
38	Which of the following key point(s) is for environmental Biotechnology interventions?	**All of the above**
39	Biological systems mat be used to clean pollutants,	**It should be highly toxic**
40	Optimum PH for local environmental conditions lies in the range of	**6.5 -7.5**
41	Why sands and gravels are best for land-based applications?	**Because sands and gravels are more porous in nature**
42	What is white Biotechnology?	**Both industrial and Environmental Biotechnology**
43	A use of thermophilic enzymes is seen in the application of pfu polymerase for PCR. Pfu stands for	**Pyrococcus furiosus**

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44	Which of the following environmental parameter has no effect	**Presence of other contaminants in the soil**
45	An extremophile	**Live in extreme conditions**
46	Ralstonia eutropha, contains which compound in its backbone?	**Sulphur**
47	DNA polymerase called Taq polymerase was derived from	**Thermus Aquaticus**
48	Which bacterium have made possible the widespread use of a revolutionary technology	**Thermus Aquaticus**
49	Those species which live in higher KCL levels are known as	**Halophiles**
50	The average oceanic temperature is around	**1 – 3 C**
51	Environmental Biotechnology involves	**All of the above**
52	Extremophiles, having a cytoplasm that contain a higher solute concentration than is present in their surrounding are	**Halophiles**
53	Which of the following is the most abundant biopolymer on this planet, as well as major product of photosynthesis?	**Cellulose**
54	The temperatures tolerated by psychrophiles are less then	**15 C or lower**
55	Generally, the local environmental conditions required would ideally after a temperature of	**20- 30 c**
56	Which of the following group is now recognized forming a distinct evolutionary line?	**Archaebacteria**
57	Multicellular organism cannot tolerate temperature_____	**60 c**

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